

Patrick O'Brien

Identity & Access Management | Cloud Automation | Microsoft Entra ID / Azure

Baltimore, MD | patobrien3590@gmail.com

pobriendev.github.io/portfolio | github.com/pobriendev | linkedin.com/in/patrick-o-brien-658125257

PROFESSIONAL SUMMARY

Identity and access automation engineer with a B.S. in Computer Science (2025), currently administering Microsoft Entra ID, Microsoft 365, and Exchange for 400+ users across 100+ properties. Builds production Python tooling on the Microsoft Graph API automating the full account lifecycle — provisioning, licensing, group membership, and deprovisioning — with least-privilege authentication and full audit logging. Software engineering background in REST API design, data modeling, and CI/CD applied to identity problems.

SKILLS

Identity & Cloud: Microsoft Entra ID (Azure AD), Microsoft Graph API, Microsoft 365, Exchange Online, Azure, Azure Key Vault, user lifecycle management (JML), role-based access control, least-privilege access, OIDC workload identity federation, audit logging

Automation & Scripting: Python, PowerShell, REST API integration, CLI tooling, YAML configuration, scheduled/unattended workflows

DevOps & Tooling: Terraform, infrastructure as code, Git/GitHub, GitHub Actions (CI/CD), Docker, pytest, automated testing

Software Engineering: TypeScript, JavaScript, SQL, Java, C++, FastAPI, React, SQLAlchemy, PostgreSQL, MySQL, JWT auth

EXPERIENCE

IT Support Specialist, Software Automation Focus

Jun 2025 – Present

Habitat America

Annapolis, MD

- Built a Python automation tool on the Microsoft Graph API that fully automates Entra ID user provisioning and deprovisioning across 100+ properties and 400+ users, replacing manual per-account setup with a single unattended workflow.
- Administer Microsoft Entra ID, Microsoft 365, and Exchange environments for 400+ users, covering account lifecycle, license assignment, group membership, and access permissions across 100+ managed properties.
- Designed app-only (client-credentials) Graph API authentication scoped to least-privilege permissions, with dry-run validation and an audit log of every write, so identity changes are reviewable and reversible.
- Diagnose and resolve system and application issues under time constraints across a distributed property portfolio.

Database Administrator

Mar 2019 – May 2025

Mike's Mowing (family business)

Ridgely, MD

- Independently designed, built, and maintained a Microsoft Access database covering customer records, service history, and scheduling, owning data integrity and schema structure over a 6-year period while working full-time alongside coursework.

Job Coach

Mar 2013 – Mar 2019

Benedictine — On-the-job coaching and support for individuals with disabilities in workplace settings.

Ridgely, MD

PROJECTS

Entra ID Infrastructure as Code

github.com/pobriendev/entra-terraform

Terraform, Azure, Microsoft Entra ID, Azure Key Vault, GitHub Actions (OIDC)

- Provisioned the Entra ID infrastructure backing my Graph API automation tools as Terraform: app registration, least-privilege Graph application permissions, service principal role assignments, and Key Vault secret storage with 180-day rotation.
- Solved the remote-state bootstrap problem with a separate root module, locking the state account down as a credential store: Entra-only auth, account keys disabled, scoped data-plane role, versioning and soft delete.
- Built a plan-on-PR pipeline authenticating via OIDC workload identity federation with no stored secret, using a read-only CI identity and redacted plan output for a public repository.

Employee Provisioning Tool

github.com/pobriendev/employee-provisioning-tool

Python, Microsoft Graph API, Entra ID, YAML

- Python CLI automating Entra ID onboarding and offboarding end-to-end: account creation or role-account reuse, license assignment, group membership, and credential delivery, replacing a manual portal process.
- App-only authentication scoped to least-privilege Graph permissions, with dry-run mode, lockout-first offboarding, and a full audit log for every write action.

Entra Stale Accounts

github.com/pobriendev/entra-stale-accounts

Python, Microsoft Graph API, PyPI, GitHub Actions

- Published a read-only CLI to PyPI (pip install entra-stale-accounts) flagging inactive Entra ID accounts past a configurable threshold, including never-signed-in accounts — an access-hygiene and audit tool.
- Generalized from production account-lifecycle automation into a standalone package with a mocked test suite, CI via GitHub Actions, and automatic Graph API retry/backoff.

Ticket Management System

github.com/pobriendev/ticket-system

FastAPI, PostgreSQL, React, JWT, Docker, GitHub Actions

- Helpdesk ITSM application with role-based access control, query-level visibility scoping, an enforced ticket status lifecycle, priority-derived SLA due dates, and an automatic audit trail on every change.
- 175+ backend tests (~96% coverage) and 125+ frontend tests; CI verifies migrations match the models, boots the Docker image, and is the only path to production. Live: ticket-system-azure-omega.vercel.app.

EDUCATION

Delaware State University — B.S. in Computer Science, GPA 3.57

2025

Delaware Technical Community College — A.S. in Computing and Information Science

2020